

RESEARCH REPORT

Search and Matching in the Age of AI: Algorithmic Disruption in the Professional-Services Labour Sub-Market

An analysis of modern hiring techniques

By **Joe Harding**

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1 Abstract

I develop a partial-equilibrium search-and-matching model of a professional-services labour sub-market in order to ask how artificial intelligence is reshaping its core mechanics: the technology that matches workers to workplaces, the surplus a completed match generates, and the Nash Bargaining rule by which that surplus is divided. It uses the Diamond–Mortensen–Pissarides (DMP) framework, where we derive formal comparative statics for labour-market tightness and wages with respect to AI-induced shocks to matching efficiency, match surplus and recruiting cost, and then evaluates the resulting equilibrium against the Hosios (1990) efficiency benchmark. The central analytical result is a separation theorem***: shocks to matching efficiency, match surplus and recruiting cost move the decentralised and socially efficient equilibria in parallel, so no shock can, themselves, make the professional-services sub-market more or less efficient. Only a change in workers’ bargaining power and the matching function’s elasticity can move the market towards social efficiency, and there is a economically coherent, so far untested, reason to think AI-mediated hiring is moving that wedge. The report closes with five testable hypotheses linking the model to recent evidence on AI-mediated hiring.

***Note: The predictions of the separation theorem arise from the assumptions of the DMP model rather than representing a general result about AI and labour markets.

Search and matching theory treats unemployment as the outcome of a costly process of finding a suitable job, rather than as a simple imbalance of labour supply and demand (Diamond, 1982; Mortensen and Pissarides, 1994; Pissarides, 2000) and such markets do not clear. The matching function, the technology that turns a pool of job-seekers and a pool of vacancies into hires, is normally treated as a stable, slow-moving feature of a labour market summarising institutional factors.

The latter assumption is harder to defend in professional services today, because hiring in this market is now mediated by algorithms on both sides of the market simultaneously. On the firm side, applicant-tracking systems (ATS) and AI interviewers screen candidates: one industry survey reports that roughly 63pct of US job-seekers have already been interviewed by an AI system. (Greenhouse, reported in Fortune, 2026). On the worker side, generative AI is now routinely used to draft and tailor applications, with survey evidence suggesting that over a third of job-seekers have used AI tools somewhere in their application process (Resume Genius, 2026). Several industry reports describe the joint result as a ”signal-to-noise” problem: application volumes rise even as their average informativeness falls, prompting firms to add further layers of automated screening in response (leadgain, 2026).

2 Introduction

Three distinct literatures exist with respect to our question, and it is worth being precise about where this report sits relative to each. A first, substantial literature treats AI as a shock to the production technology underlying specific tasks, following the framework of Acemoglu and Restrepo (2018); recent applications to US occupation-level data (Hampole, Papanikolaou, Schmidt and Seegmiller, 2025) and UK firm-level employment (Klein Teeselink, King’s College London) document how AI exposure changes labour demand, typically holding the technology by which workers and firms find one another fixed. A second literature, the canonical DMP search-and-matching framework, does the opposite: it treats the matching technology as the central object, but as exogenous, an assumption that was defensible when search costs were dominated by physical and informational frictions evolving on a timescale of years, and is harder to defend when search is conducted substantially through instantaneous software. A third literature documents specific empirical failures of algorithmic hiring directly. A joint Harvard Business School–Accenture study surveying more than 8,000 workers rejected by ATS and over 2,250 executives across the US, UK and Germany found that 88pct of employers acknowledged their ATS routinely screens out qualified candidates whose applications do not precisely match rigid keyword or pre-established criteria (Fuller and Raman, 2021). This literature documents the failure mode in detail but does not embed it in an equilibrium framework capable of speaking to wages, vacancy creation, or welfare.

This report positions itself at the intersection of the first two. It treats AI as a shock to the surpluses generated by a completed match: the production-side channel emphasised by the first literature, and to the technology and bargaining environment through which that match is found in the first place, the object emphasised by the second. It then borrows from the second literature, specifically the Hosios efficiency condition, to ask a question neither the first nor the third literature is set up to answer: whether AI-mediated hiring outcomes, even when they are privately rational for both firms and workers individually, are socially efficient. We find a clean separation between channels that move outcomes and the single channel that moves efficiency, is, to my knowledge, not present elsewhere in the applied AI-and-labour literature, which has generally not combined a task-exposure framework with a constrained-efficiency benchmark.

The sections proceed as follows. Section 3 motivates the choice of a single, professional-services sub-market as the object of analysis. Section 4 sets out the DMP model and derives its comparative statics formally. Section 5 introduces the Hosios efficiency bench-

mark and proves the separation result summarised above. Section 6 works through five distinct channels by which AI plausibly disrupts this sub-market, several of them new or substantially sharpened relative to the existing literature. Section 7 discusses the model's scope conditions and the identification problems any empirical test of it would face. Section 8 sets out five testable hypotheses, and Section 9 concludes.

3 The Professional-Services Sub-Market

Search-and-matching models are typically applied separately to distinct submarkets, groups of workers and jobs that compete with one another but not directly with workers and jobs elsewhere in the economy (Pissarides, 2000, ch. 1). This report applies the model to a single submarket: professional services and white-collar occupations, roles in finance, consulting, law, administration and analytics that are knowledge intensive and recruited for primarily online (online job boards, ATS-managed applications, LinkedIn).

Two features make this submarket the right setting for studying AI disruption of search and matching. First, recruitment in this submarket is highly digitised: sourcing, screening and interviewing are conducted substantially online, which is precisely where AI tools, on both the applicant and the recruiter side, operate. Second, this submarket is independently the one most exposed to AI task substitution, which changes the value of a successful match along with the probability of matching. Firm-level evidence finds that employment at AI-exposed firms fell by an average of 4.5pct, with the decline concentrated almost entirely in junior positions, which fell by 5.8pct (Klein Teeselink, King's College London). US evidence over 2010–2023 similarly finds that occupations with higher AI exposure subsequently see reduced labour demand, although firm-level productivity gains offset the loss of labour (Hampole, Papanikolaou, Schmidt and Seegmiller, 2025).

The model below is accordingly a partial-equilibrium model of this one submarket: wages and hiring conditions elsewhere in the economy are taken as given, and the object of interest is how AI changes unemployment, vacancies and wages within it. Section 6 returns to what this restriction costs the analysis. A general equilibrium model with cross-sectional mobility across several unique job markets would be more representative of the labour market as a whole, but the following sections would lose intuition and predictive power.

4 The Canonical Search-and-Matching Model

4.1 Matching Technology and Labour-Market Tightness

Let u denote the unemployment rate and v the vacancy rate in the submarket. Matches form according to a constant-returns, Cobb-Douglas style, matching function

$$x(u, v) = m u^\xi v^{1-\xi}, \quad 0 < \xi < 1, \quad (1)$$

where $m > 0$ is matching efficiency. Define tightness $\theta \equiv v/u$. Two hazard rates follow directly: the rate at which a vacancy is filled,

$$q(\theta) = \frac{x(u, v)}{v} = m \theta^{-\xi}, \quad (2)$$

and the rate at which an unemployed worker finds a job,

$$f(\theta) = \frac{x(u, v)}{u} = m \theta^{1-\xi}. \quad (3)$$

A tighter market means vacancies are harder to fill but jobs are easier to find since $f'(\theta) > 0$ and $q'(\theta) < 0$. This trade-off is elaborated on in the following chapters. These may be interpreted as the probabilities that a vacancy is filled, $q(\theta)$, and the probability that a worker finds a job, $f(\theta)$, in a given period.

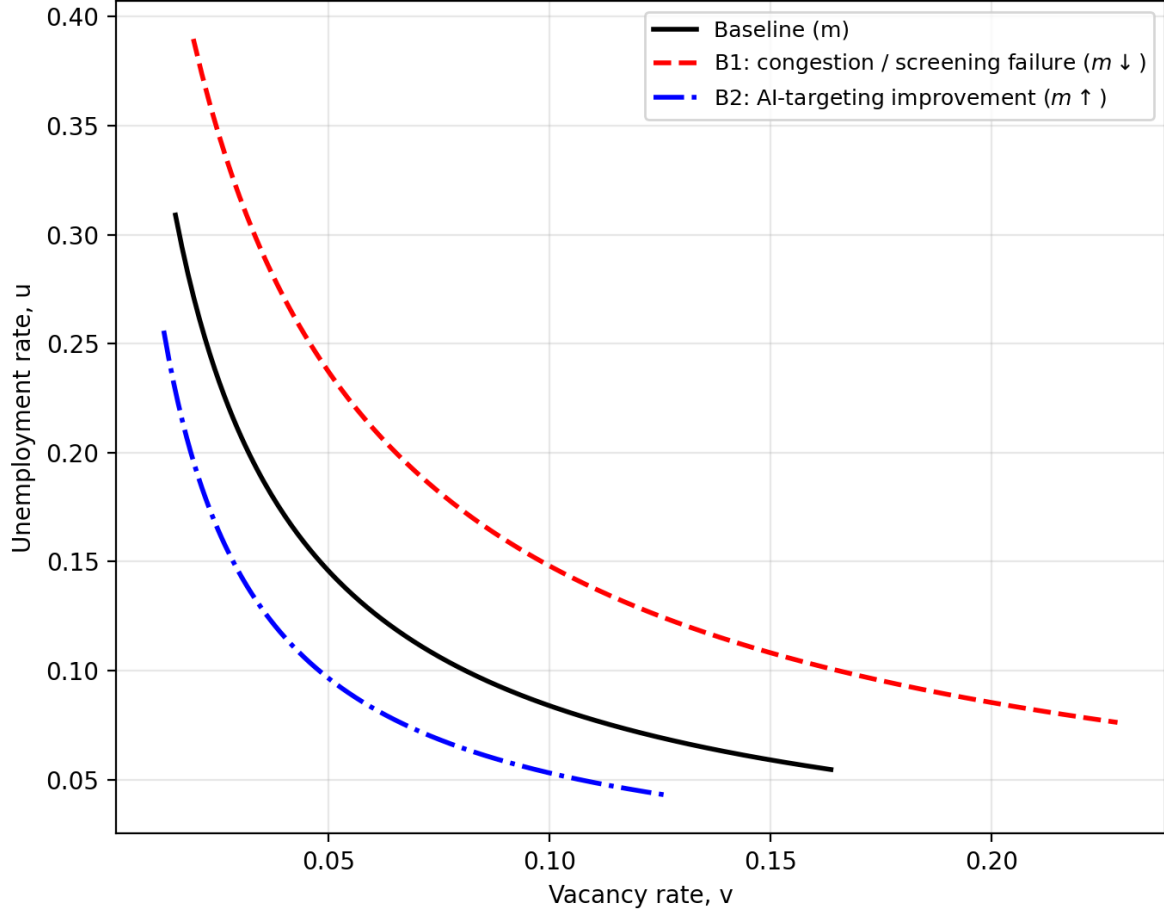
4.2 Flows and the Beveridge Curve

Employed workers separate from their job at an exogenous rate λ . In steady state, separations equal hires:

$$\lambda(1 - u) = f(\theta) u. \quad (4)$$

Equation (4) is the Beveridge curve: for given m and λ , it traces the (u, v) pairs consistent with steady-state flow balance, and a fall in m shifts it outward. Section 6.1 returns to this relationship once the AI-specific shocks to m are introduced, and Figure 1 there shows the resulting shifts for this submarket directly.

Figure 1: Beveridge Curve Shifts Under AI-Driven Matching-Efficiency Channels (Section 5.1)



4.3 Job Creation: Firm Value Functions and Free Entry

Let r be the discount rate, c the cost for having a vacancy open, y the output, and w the wage. The values of a vacancy, V , and a filled job, J , satisfy

$$rV = -c + q(\theta)(J - V), \quad rJ = y - w - \lambda(J - V). \quad (5)$$

Free entry gives $V = 0$, which gives $J = c/q(\theta)$ and, substituted into the second equation, the job creation (JC) condition:

$$\frac{c}{q(\theta)} = \frac{y - w}{r + \lambda}. \quad (6)$$

For given y , (6) defines a downward-sloping relationship between θ and w since a higher wage leaves less surplus to cover vacancy costs, so fewer firms post vacancies.

4.4 Wage Determination: Nash Bargaining

Let U and E be the values of unemployment and employment, with flow value of non-work z :

$$rU = z + f(\theta)(E - U), \quad rE = w + \lambda(U - E). \quad (7)$$

These equations imply that the value of non-work is equal to the sum of the current payoff e.g. leisure and unemployment benefits and the future possibility of employment. Likewise, the payoff to the worker is the wage plus the probability and payoff of becoming unemployed in the given period, the latter part is often negative since we expect $E \geq U$. The wage is set by Nash bargaining over the match surplus with worker bargaining power $\beta \in (0, 1)$. Combining the sharing rule $(E - U) = \beta[(E - U) + (J - V)]$ with (5)–(7) and free entry gives the wage-setting (WS) curve:

$$w = (1 - \beta)z + \beta(y + c\theta). \quad (8)$$

Equation (8) is upward-sloping in θ : a tighter market improves a worker's opportunity cost, raising the bargained wage.

4.5 Equilibrium and Comparative Statics

Substituting (8) into (6) and rearranging gives a single equation that implicitly defines equilibrium tightness θ :

$$c \left[\frac{(r + \lambda)\theta^\xi}{m} + \beta\theta \right] = (1 - \beta)(y - z). \quad (9)$$

Lemma 1 (Comparative statics of tightness). Assume $y > z$, so a viable match generates strictly positive surplus over the value of non-work. The equilibrium tightness θ defined by (9) satisfies $\partial\theta/\partial m > 0$, $\partial\theta/\partial y > 0$, and $\partial\theta/\partial c < 0$.

- $\partial\theta/\partial m > 0$
 - An increase in matching efficiency, m , corresponds with an increase in equilibrium labour market tightness, θ , as firms create more vacancies when matches are formed easily.
- $\partial\theta/\partial y > 0$
 - An increase in worker productivity, y , increases equilibrium labour market tightness, θ , by making vacancy creation more profitable.
- $\partial\theta/\partial c < 0$

- An increase in recruiting costs, c , reduces equilibrium labour market tightness, θ , as higher hiring costs discourage firms from creating vacancies.

Proof sketch. Write (9) as

$$G(\theta) \equiv c(r + \lambda)\theta^\xi/m + c\beta\theta - (1 - \beta)(y - z) = 0.$$

Since

$$\partial G/\partial \theta = c(r + \lambda)\xi\theta^{\xi-1}/m + c\beta > 0$$

for all $\theta > 0$, the implicit function theorem gives

$$\partial \theta/\partial x = -(\partial G/\partial x)/(\partial G/\partial \theta)$$

for each parameter x . Direct differentiation gives

$$\partial G/\partial m = -c(r + \lambda)\theta^\xi/m^2 < 0, \partial G/\partial y = -(1 - \beta) < 0,$$

and

$$\partial G/\partial c = (r + \lambda)\theta^\xi/m + \beta\theta > 0,$$

which shows the inequalities of these first order partials match our assumption. ■

Corollary 1 (Wage comparative statics). Substituting $\theta(\cdot)$ into (8) gives:

$$\partial w/\partial m = \beta c \cdot \partial \theta/\partial m > 0$$

and

$$\partial w/\partial y = \beta(1 + c \cdot \partial \theta/\partial y) > 0.$$

The sign of

$$\partial w/\partial c = \beta\theta + \beta c \cdot \partial \theta/\partial c$$

is, by contrast, ambiguous in general. It is positive only if the direct effect of c on the worker's share of the firm's recruiting cost saving outweighs the indirect effect of c on tightness.

These results put the diagrammatic comparative statics used later in the chapter on a formal footing, and the ambiguity flagged in Corollary 1 is itself informative: not every AI-driven cost shock to this submarket has a clean theoretical prediction for wages, and §6.2 and §7 are written to reflect that.

5 Efficiency Benchmark: The Hosios Condition

Digression: Hosios Condition

In a search-and-matching model, agents actions carry an external effect on the all other agents within the market that are not accounted for in their own actions. This create search externalities. Hosios (1990) showed that an equilibrium is socially efficient if $\beta = \xi$ where β is the worker bargaining power and ξ is the elasticity of the matching function with respect to vacancies (how strongly additional vacancies increase the number of matches).

Lemma 1 tells us what AI does to outcomes but does not say if these outcomes are efficient. A social planner choosing θ directly, acknowledges the search externalities that private agents do not and solves a problem that maximises output net of recruiting costs subject to equation (4). The resulting optimality condition takes a form akin to equation (9):

$$c \left[\frac{(r + \lambda) \theta^\xi}{m} + \xi \theta \right] = (1 - \xi)(y - z), \quad (10)$$

which is the decentralised condition (9) where $\beta = \xi$ is substituted to find social efficiency (Hosios, 1990; Pissarides, 2000). The intuition follows from two externalities embedded in any matching market with constant-returns search technology (since the function is Cobb-Douglas the CRTS property is inherited in scope of returns to matching). An additional job-seeker congests the market for other seekers, raising u lowers the job-finding rate, $f(\theta)$, facing everyone else at given v , while simultaneously raising the chance any given vacancy is filled. An additional vacancy generates the mirror-image pair of effects. The elasticity ξ in (1) measures the searchers' aggregate marginal social contribution to matches; the bargaining weight β measures the share of the realised surplus workers are able to privately capture after negotiation. These are independent, fixed by different mechanisms, one technological, one institutional, so there is no general reason for them to coincide.

Proposition 1 (Hosios efficiency). Let θ^{eff} solve equation (10). Further let be $\theta(k)$ as the solution to $c[(r + \lambda)\theta^\xi/m + k\theta] = (1 - k)(y - z)$ for $k \in \{\beta, \xi\}$, so that $\theta(\beta)$ is the decentralised equilibrium and $\theta(\xi) = \theta^{eff}$. Then $\theta(\beta) = \theta^{eff}$ if and only if $\beta = \xi$; more generally, $\theta(k)$ is strictly decreasing in k , so $\beta > \xi$ implies $\theta < \theta^{eff}$ (inefficiently low vacancies, inefficiently high unemployment) and $\beta < \xi$ implies $\theta > \theta^{eff}$ (inefficiently high vacancy creation, an inefficiently tight market).

Proof sketch. Define $H(\theta; k) \equiv c(r + \lambda)\theta^\xi/m + ck\theta - (1 - k)(y - z)$. As in Lemma

1, $\partial H/\partial \theta > 0$; in addition $\partial H/\partial k = c\theta + (y - z) > 0$ given $\theta > 0$ and $y > z$. The implicit function theorem gives $d\theta(k)/dk = -(\partial H/\partial k)/(\partial H/\partial \theta) < 0$. Since $\theta(k)$ is strictly decreasing, $\beta > \xi$ implies $\theta(\beta) < \theta(\xi) = \theta^{eff}$, and conversely. ■

Corollary 2 (Separation of levels and efficiency). Equations (9) and (10) are identical and differ only in the parameter, β or ξ . Every comparative static in Lemma 1 therefore applies to θ^{eff} as well as θ : $\partial \theta^{eff}/\partial m > 0$, $\partial \theta^{eff}/\partial y > 0$, $\partial \theta^{eff}/\partial c < 0$, by the same proof. Consequently, none of the matching-efficiency, match-surplus or recruiting-cost channels examined in §6.1 or in §6.4 can, on their own, narrow or widen the gap between the decentralised and efficient equilibrium, each shifts θ and θ^{eff} in the same direction. It is exclusively the wedge between β and ξ , examined in §6.5, that determines whether AI moves this submarket toward or away from constrained efficiency.

Figure 2: Decentralised vs. Efficient Equilibrium
(Proposition 1: $\beta > \xi$ case)

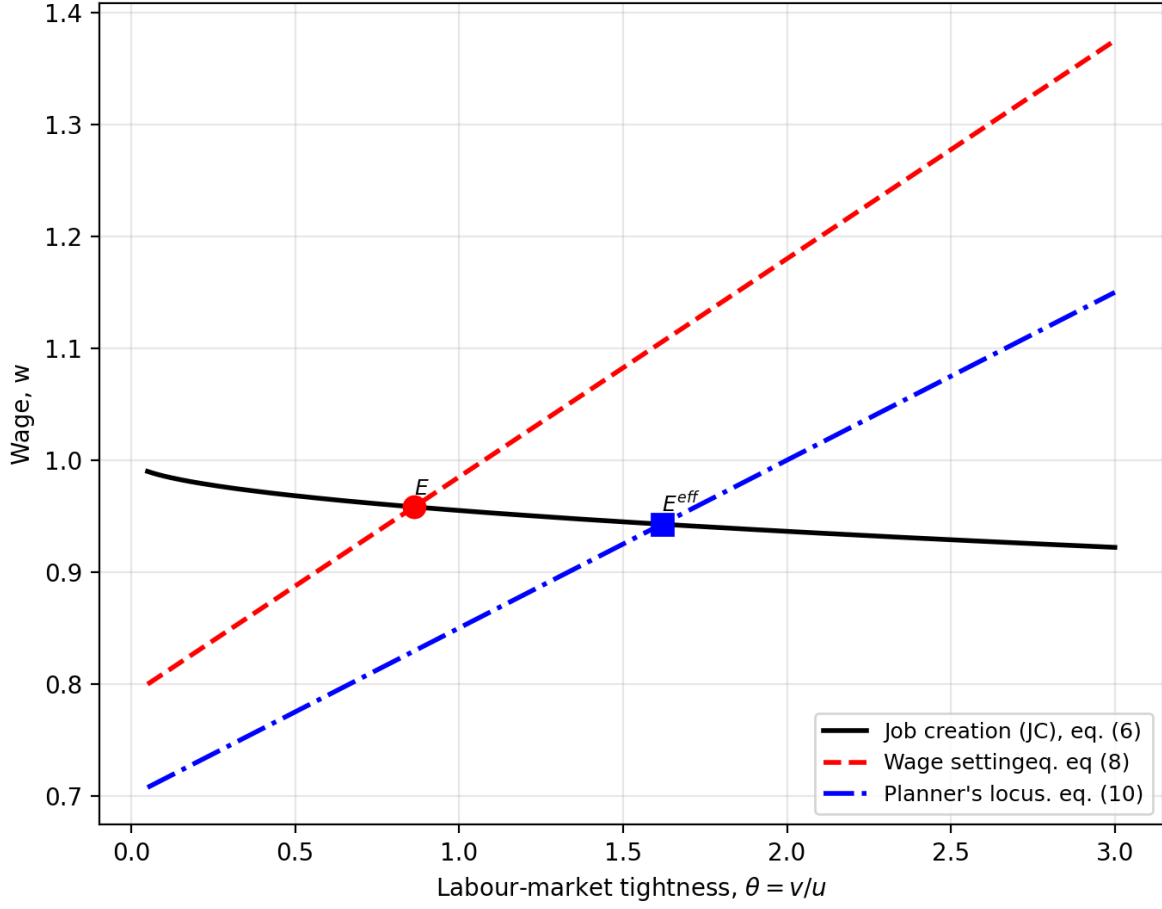


Figure 1: Figure 2 illustrates Proposition 1 numerically, holding m , y and c fixed (Corollary 2) and varying only the workers bargaining parameter between $\beta = 0.65$ and $\xi = 0.50$, an illustrative case in which workers' bargaining weight exceeds the matching elasticity. The job-creation curve is unchanged across the two equilibria by construction, and the decentralised equilibrium E sits at lower tightness than the planner's E^{eff} , exactly as Proposition 1 predicts.

Note: These are arbitrary figures that may not necessarily reflect data but illustrate the expected outcomes.

6 AI as a Disruption to Search, Matching and Bargaining

6.1 Matching Efficiency: Congestion versus Screening-Accuracy Channels

A first pass at this submarket might treat AI’s effect on matching efficiency as a single, ambiguously-signed change. AI helps matching in some ways and hurts it in others, and the net effect is a question for later econometric analysis. That is true, but it conflates two channels that are conceptually distinct and, may be moving in different directions for different reasons.

The congestion channel operates through the sheer volume of search activity entering the matching function. Generative AI has lowered the marginal cost of producing a tailored application, and job-seekers have responded by applying more: over a third of jobseekers surveyed report using AI tools somewhere in their application process (Resume Genius, 2026), and industry analysts describe the resulting environment as a ‘signal-to-noise crisis’ in which application volumes rise even as their average informativeness falls (leadgain, 2026). *Ceteris-paribus*, a pure rise in search-side volume of this kind is formally equivalent to a fall in the level term m : more applications must be processed to generate the same number of hires, and firms have responded by adding further automated screening, explicitly to cut through application autonomously (Greenhouse, reported in Fortune, 2026).

The screening-accuracy channel is distinct: it operates through the probability that a genuinely viable match, is correctly identified by automated screening, independent of how many competing applications exist. Fuller and Raman’s joint Harvard Business School–Accenture study, surveying more than 8,000 workers rejected by automated hiring systems and over 2,250 executives across the US, UK and Germany, found that 88pct of employers acknowledged their ATS routinely exclude qualified candidates whose resumes do not precisely match narrow keyword, credential or continuous-employment criteria, a population we crudely extrapolate to be 27 million workers in the US labour market alone (Fuller and Raman, 2021). In the notation of §4, this is a Type-II error: a worker–vacancy pair with positive surplus, $y - z > 0$, exists but is never realised, not because too many applications are competing for attention, but because the screening rule itself is wrong.

A pure congestion effect is addressable by scaling screening capacity without altering the underlying decision rule. A screening-accuracy effect is not, since it reflects a property of the rule set by recruitment teams: the rigid keyword and credential matching

criteria documented by Fuller and Raman scale perfectly, rejecting ten million applications costs an ATS no more than rejecting ten thousand, while remaining systematically wrong. Where the AI-assisted targeting described by some industry sources, which can read an entire CV, infer skill adjacencies, and surface roles a keyword search would miss (Frontline Source Group, 2026), represents a genuine improvement in screening accuracy, the Fuller and Raman evidence represents the opposite: a screening technology that is fast and scalable but is frequently inaccurate. Figure 1 summarises both possibilities on the Beveridge curve introduced in §4.2: B_1 depicts an outward shift consistent with either the congestion channel or the screening accuracy failure documented above, and B_2 depicts the inward shift implied by genuine targeting improvements. Which dominates in practice is an empirical question (see Hypothesis 1 in §8); what the decomposition adds is two separately falsifiable channels in place of one unsigned one.

By Corollary 2, it is worth noting immediately that neither sub-channel, however large, can by itself make this submarket more or less efficient: both operate purely through m , which shifts θ and θ^{eff} together.

6.2 The Match Surplus and Recruiting Cost: The Job-Creation Channel

Independently of whether AI helps or hinders the matching process, it changes the value of the thing being matched. Where AI substitutes directly for the codifiable, narrowly-scoped tasks that make up much of an entry-level professional role, the surplus a firm earns from hiring a human falls: y in (6) is lower for these roles than for roles built around judgement, client relationships or supervision, which AI is more likely to complement than replace. This is consistent with the evidence cited in §3: aggregate employment effects at AI-exposed UK firms are concentrated almost entirely in junior positions, while senior roles, and average pay at exposed firms, are comparatively unaffected (Klein Teeselink, King’s College London). By Lemma 1 and Corollary 1, a fall in y for these roles lowers both θ and w unambiguously.

A complementary mechanism works through c rather than y : maintaining an AI screening layer, licensing ATS software, building and validating automated interview tools, is itself a vacancy cost. By Lemma 1, a rise in c depresses θ in the same direction as a fall in y , so the two channels reinforce one another on tightness; by Corollary 1 their wage implications need not reinforce one another, since the sign of $\partial w / \partial c$ depends on the relative magnitude of a direct cost-sharing effect and an indirect tightness effect. This is one reason §8 treats wage predictions for the c -channel more cautiously than for the y -channel.

By Corollary 2, neither the y nor the c channel can move this submarket toward or away from efficiency on its own, whatever their magnitude: both shift θ and θ^{eff} in parallel.

6.3 Ghost Vacancies and the Measurement of v

The Beveridge curve in §4.2 assumes v measures genuine search effort. Survey evidence complicates this for the modern professional-services submarket: 67pct of US job-seekers in one 2026 survey reported suspecting that a posting they applied to was fake, misleading, or never actually intended to be filled (Resume Genius, 2026). If a meaningful share of posted vacancies are not a genuine search effort on behalf of the firm, posted for internal benchmarking, to build a talent pipeline, or simply left live after a role is filled means that the measured v overstates true vacancy creation, and an apparent outward shift of the Beveridge curve may partly be a measurement error rather than a genuine fall in m . This is a methodological point worth carrying into any empirical design: vacancy counts scraped from job boards are not a clean proxy for v in this submarket in the way the canonical model assumes, a point §6.5 returns to from a different angle.

6.4 Duration Dependence and Algorithmic Screening

An extension of the search model allows the job-finding function to depend on how long a worker has already been unemployed, represented by τ :

$$f_i(\theta, \tau_i) = m \theta^{1-\xi} \phi(\tau_i), \quad \phi'(\tau) \leq 0. \quad (11)$$

Two results motivate $\phi'(\tau) \leq 0$: skills can depreciate during a non-employment spell, lowering a worker’s effective productivity in any future match (Pissarides, 1992), and, independently of any true productivity loss, employers appear to use unemployment duration itself as a statistical screen. A field experiment sending otherwise identical fictitious resumes found callback rates declined sharply with the length of the reported unemployment spell (Kroft, Lange and Notowidigdo, 2013).

The Fuller and Raman (2021) finding gives this channel an unusually concrete mechanism in the AI-hiring context: their survey reports that a near-majority of the firms studied, 49pct, explicitly screen out resumes showing employment gaps of six months or longer. Where Kroft, Lange and Notowidigdo document duration-based statistical discrimination as an empirical regularity, the ATS evidence shows the same mechanism operating as an explicit, automatable decision rule, meaning $\phi(\tau)$ for large τ falls through

a single parameter in a piece of widely-deployed software. This connects directly back to §6.1’s screening accuracy channel and to the junior vacancy result in §6.2: a displaced junior professional who searches for longer is increasingly disadvantaged in an AI-screened submarket relative to an otherwise identical worker who was re-employed quickly, and the disadvantage compounds through exactly the mechanism formalised in (11).

6.5 AI and the Bargaining-Matching Wedge

Sections 6.1 to 6.4 establish that AI plausibly affects m , y and c . By Corollary 2, however, none of this, however large in magnitude, bears on whether the resulting equilibrium is efficient. That question turns entirely on the wedge between β and ξ , and this section sets out, explicitly as a conjecture rather than an established result, why AI-mediated hiring might be moving that wedge specifically in this submarket. No published estimate of either parameter’s response to AI-mediated hiring presently exists (to my knowledge).

On the bargaining side, generalised Nash bargaining ties β to each party’s contestability and outside option, unemployment or not hiring, at the point of negotiation. A market in which a single vacancy reliably attracts a shortlist of qualified, near-identical applicants, implied by the congestion channel in §6.1, plausibly weakens each individual applicant’s bargaining position because the firm’s credible threat of continued search is less costly to execute. If so, AI-mediated hiring should be associated with a fall in β in this submarket specifically.

On the matching-elasticity side, ξ measures how much an additional searcher congests the process for other searchers, holding vacancies fixed. Automated screening that processes large applicant pools at near-zero marginal cost weakens the technological link between applicant volume and processing delay that, under human review, would normally underpin a non-trivial ξ , which would push ξ down alongside β . But the screening-accuracy failures documented by Fuller and Raman (2021) are at least consistent with the opposite possibility, that automated systems become noisier and marginally slower as volume rises, which would keep ξ from falling as fast, or push it the other way.

The value of Proposition 1 is that its qualitative implications are sharp even without resolving this ambiguity empirically. If β falls by more than ξ , the case illustrated numerically in Figure 3, where an initial $\beta > \xi$ configuration is pushed further apart, the wedge widens and the decentralised equilibrium moves further below the planner’s efficient tightness, consistent with under-provision of genuine vacancies and elevated long-term joblessness among displaced professional-services workers. If instead β falls far enough to flip the

ranking to $\beta < \xi$, Proposition 1 predicts the opposite: an inefficiently over-tight market, with an excess of vacancy creation relative to the social optimum. This second case offers a structural complement to the purely measurement-based account of §6.3 where the high volume of low-quality postings is potentially a genuine equilibrium symptom of firms over-posting because the private cost of doing so has fallen below its social cost. Section 8 sets out how the two cases can be distinguished empirically without directly estimating β or ξ .

7 Discussion: Scope Conditions and Limitations

The model is narrow, and four limitations follow directly from that choice. First, it is a partial-equilibrium analysis: z and conditions in adjacent submarkets are held fixed, so workers displaced from professional services cannot be re-absorbed elsewhere in a way that feeds back on this submarket’s own outside option or tightness. A general-equilibrium treatment would let exactly this kind of cross-market linkage operate, at the cost of the brevity that makes Proposition 1 provable in closed form first order conditions.

Second, β is treated throughout as a structural parameter, but the literature is clear that it is not a directly observed one. Hagedorn and Manovskii (2008) propose calibrating β and z jointly from data on vacancy-creation costs and wage cyclicalities, precisely because the standard model is otherwise highly sensitive to assumptions about their values, a sensitivity that became salient after Shimer’s (2005) demonstration that conventional calibrations of the DMP model generate far less cyclical volatility in unemployment and vacancies than the data show. If there is no settled method for measuring β under ordinary conditions, there is certainly none yet for measuring how it responds to a new hiring technology, and the argument in §6.5 should be read accordingly: a theoretically motivated conjecture, not a calibrated result as for much of this report.

Third, the Cobb-Douglas, constant-returns matching function in (1) is the standard workhorse and well supported empirically from Petrongolo and Pissarides (2001). Cross-country evidence find the constant-returns Cobb-Douglas specification performs well across a range of settings, but a more general functional form, allowing ξ itself to vary with θ or with the mix of AI-assisted and unassisted search within u , would be needed to formalise the §6.1 decomposition as a single object rather than two competing comparative-statics layered on top of a fixed-elasticity function.

Fourth, the empirical hypotheses in §8 face an identification problem common to this literature: firm-level AI adoption is not exogenous. Firms that adopt AI screening differ systematically, in size, growth rate, sector or managerial quality, from those that do not.

Hence a naive comparison of adopting and non-adopting firms’ hiring outcomes would conflate the causal effect of AI with selection into adoption and bias estimated coefficients without controlling for, often unobservable, characteristics. Where possible, the hypotheses below are framed around ex-ante occupational or task-level exposure measures to mitigate this. A related point concerns the survey evidence motivating much of §6: the application-inflation, ghost-vacancy and AI-interview statistics cited above are self-reported perceptions and should be read as noisy suggestions of the channels they motivate rather than as direct estimates of m , v .

8 Testable Hypotheses For Later Works

Hypothesis 1 (Net matching efficiency). Because the congestion and screening-accuracy channels in §6.1 both operate through m but have different remedies, they should be separately identifiable: time-to-fill and vacancy-duration data, conditional on application volume, should reveal a residual screening accuracy effect, callback rates for observably similar, qualified candidates should be lower at firms using rigid, rule-based ATS criteria than at firms using more adaptive matching tools. This isolates the Fuller and Raman (2021) mechanism from pure congestion and is directly testable with firm-level hiring-funnel data.

Hypothesis 2 (Junior–senior composition). By Lemma 1, a fall in y for codifiable, entry-level tasks should produce a larger fall in net hiring for junior, task codifiable roles than for senior roles within the same AI-exposed firm. This is testable with firm-level vacancy and headcount data, as in Klein Teeselink’s UK evidence, and ideally instrumented with an ex-ante, occupation-level exposure measure rather than realised firm adoption, to address the selection concern raised in §7.

Hypothesis 3 (Ghost-vacancy gap). If §6.3’s measurement story is empirically important, the gap between posted vacancies and verified hiring intent, fill rates, applicant-to-offer ratios, should be systematically wider on heavily AI-mediated job boards than in referral-based channels.

Hypothesis 4 (Duration dependence under automation). Callback or interview-invitation rates should decline more steeply with unemployment duration in occupations using automated, rule based screening than in otherwise comparable occupations relying on human recruiters, controlling for occupational AI exposure, a direct test of whether (11)’s $\phi(\tau)$ falls faster under the specific mechanism documented by Fuller and Raman (2021).

Hypothesis 5 (The bargaining-matching wedge). The two cases distinguished in §6.5 are empirically separable without estimating β or ξ directly. If the wedge has moved toward $\beta < \xi$, the model predicts an excess of vacancy creation relative to the planner’s benchmark, which should show up as elevated and rising vacancy-to-hire ratios alongside the high suspected-fake-posting share documented in survey data (§5.3). The competing case, a widening $\beta > \xi$ wedge, predicts the opposite: falling vacancy-to-hire ratios alongside rising long-term unemployment among displaced professional-services workers. Because these two cases generate opposite predictions for an observable, administrative quantity, the vacancy-to-hire ratio, testing between them does not require resolving the underlying ambiguity about β and ξ separately, only signing their net effect.

9 Conclusion

AI is changing how workers and firms find each other in the first place. The contribution of this report is to show that these two questions, “what does AI do to hiring outcomes” and “does AI make this labour market more or less efficient”, are not the same question and therefore do not have the same answer. Five of the six positive channels examined here, targeting, congestion, screening-accuracy failure, task substitution, and recruiting cost, move tightness, unemployment and wages in well-defined directions, but by Corollary 2 none of them, individually or combined, can move this submarket toward or away from social efficiency from the Helois condition. Only a shift in the wedge between workers’ bargaining power and the matching function’s elasticity can do that, and the available evidence, while suggestive, does not yet pin down which way that wedge is moving. A full account of the modern professional-services labour market needs both halves of this distinction: the well-documented, first-order channels through which AI is reshaping who gets hired, and the considerably less well-documented, second-order channel that determines whether the resulting allocation is one a planner would choose.

9.1 References

Acemoglu, D. and Restrepo, P. (2018). “Artificial Intelligence, Automation, and Work.” NBER Working Paper No. 24196.

Diamond, P. (1982). “Wage Determination and Efficiency in Search Equilibrium.” *Review of Economic Studies*, 49(2), 217–227.

Fuller, J. and Raman, M., with Sage-Gavin, E. and Hines, K. (2021). “Hidden Workers: Untapped Talent.” Harvard Business School Project on Managing the Future of

Work, in collaboration with Accenture.

Hagedorn, M. and Manovskii, I. (2008). "The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited." *American Economic Review*, 98(4), 1692–1706.

Hampole, M., Papanikolaou, D., Schmidt, L.D.W. and Seegmiller, B. (2025, revised Sept. 2025). "Artificial Intelligence and the Labor Market." NBER Working Paper No. 33509.

Hosios, A.J. (1990). "On the Efficiency of Matching and Related Models of Search and Unemployment." *Review of Economic Studies*, 57(2), 279–298.

Klein Teeselink, B. King's College London, reported in "New study reveals early impact of AI on job market in UK," King's College London News, 2026.

Klein Teeselink, B. and Carey, D. King's College London / AI Objectives Institute, reported in "Study identifies key elements which determine impact of AI on jobs," King's College London News, 2026.

Kroft, K., Lange, F. and Notowidigdo, M. (2013). "Duration Dependence and Labor Market Conditions: Evidence from a Field Experiment." *Quarterly Journal of Economics*, 128(3), 1123–1167.

Mortensen, D. and Pissarides, C. (1994). "Job Creation and Job Destruction in the Theory of Unemployment." *Review of Economic Studies*, 61(3), 397–415.

Petrongolo, B. and Pissarides, C. (2001). "Looking into the Black Box: A Survey of the Matching Function." *Journal of Economic Literature*, 39(2), 390–431.

Pissarides, C. (1992). "Loss of Skill During Unemployment and the Persistence of Employment Shocks." *Quarterly Journal of Economics*, 107(4), 1371–1391.

Pissarides, C. (2000). *Equilibrium Unemployment Theory*, 2nd ed. MIT Press.

Shimer, R. (2005). "The Cyclical Behavior of Equilibrium Unemployment and Vacancies." *American Economic Review*, 95(1), 25–49.

Greenhouse, reported in "Nearly 4 in 10 job candidates have bailed on a hiring round because it required an AI interview," *Fortune*, 2026.

"AI in Recruitment 2026: The Reality on the Ground," leadgain, 2026.

"Job Search Statistics Report 2026: Trends, Challenges, & AI Data," Resume Genius, 2026.

"What are the best job search tools, AI platforms, and apps for 2026?," Frontline Source Group, 2026.